

1. When a certain car breaks down, the time that it takes to fix it (in hours) is a

random variable with the density function $f(x) = \begin{cases} ce^{-3x} & 0 \leq x \\ 0 & \text{otherwise} \end{cases}$

(a.) Calculate the value of c .

(b.) Find the probability that when this car breaks down, it takes at most 30 minutes to fix it.

2. The distribution function for the duration of a certain soap opera (in tens of hours)
3. Let X be a continuous random variable with the density function
 $f(x) = 1/4$ if $x \in (-2, 2)$
4. Let the probability density function of a random variable X be
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$$f(x) = \frac{1}{2} e^{-|x|}, \quad -\infty < x < \infty$$

Then, calculate $\text{Var}(X)$.